

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR
Deemed University
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CORNERSTONE PROJECT
ON
DESIGN OF CANTEEN MANAGEMENT SYSTEM

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CERTIFICATE

This is certified that **Surya Pratap Singh** (MCCA25O1050) has submitted the cornerstone project report titled **Design of Canteen Management System** under the mentorship of **Dr. PARUL SAXENA, Assistant Professor**, in partial fulfilment of the requirement for the award of degree of Master in Computer Application of Computer Science and Engineering from Madhav Institute of Technology and Science, Gwalior.

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ABSTRACT

Most of the colleges still operate their canteen systems the old-fashioned way. Students tend to form long lines, especially at lunch (often leading to mix-ups and delayed orders due to a rush of people). In our college canteen, we have found the same issue where many students waste time merely placing an order.

To remedy this, we designed a Design of Canteen Management System. The main function of the Design of Canteen Management system is to allow students to order quickly and easily. Rather than standing in line, students can use their smartphones and/or laptops to view menus and place orders.

Design of Canteen Management system is a web-based application where students/staff and admin have their own login access. When an order is placed through the Design of Canteen Management System, it is stored in a database and sent directly to the kitchen staff reducing mix ups and helping to manage orders.

We used Python and Django for the back-end development of the Design of Canteen Management System and utilized HTML, CSS, and JavaScript for the front-end. We also used MySQL for storing all the data used by the system. This will help save time, reduce errors, and improve the overall service within the canteen.

Keywords: Canteen Management, Web Application, Django Framework, Online Food Ordering, Digital Billing, Order Tracking, MySQL

सार

अधिकांश कॉलेज अभी भी अपने कैंटीन सिस्टम को पुराने तरीके से संचालित करते हैं। छात्र लंबी लाइनें बनाते हैं, खासकर दोपहर के भोजन पर (अक्सर लोगों की भीड़ के कारण मिक्स-अप और विलंबित ऑर्डर के लिए अग्रणी)। हमारे कॉलेज की कैंटीन में, हमें एक ही मुद्दा मिला है जहां कई छात्र केवल एक ऑर्डर देने में समय बर्बाद करते हैं।

इसका समाधान करने के लिए, हमने कैंटीन प्रबंधन प्रणाली का एक डिजाइन तैयार किया है। कैंटीन प्रबंधन प्रणाली के डिजाइन का मुख्य कार्य छात्रों को जल्दी और आसानी से ऑर्डर करने की अनुमति देना है। लाइन में खड़े होने के बजाय, छात्र मेनू देखने और ऑर्डर देने के लिए अपने स्मार्टफोन और/या लैपटॉप का उपयोग कर सकते हैं।

कैंटीन प्रबंधन प्रणाली का डिजाइन एक वेब-आधारित एप्लिकेशन है जहां छात्रों/कर्मचारियों और व्यवस्थापक के पास अपनी लॉगिन पहुंच होती है। जब कैंटीन प्रबंधन प्रणाली के डिजाइन के माध्यम से एक आदेश दिया जाता है, तो इसे एक डेटाबेस में संग्रहीत किया जाता है और सीधे रसोई के कर्मचारियों को भेजा जाता है ताकि मिश्रण को कम किया जा सके और ऑर्डर प्रबंधित करने में मदद मिल सके।

हमने कैंटीन प्रबंधन प्रणाली के डिजाइन के बैक-एंड विकास के लिए पायथन और Django का उपयोग किया और फ्रंट-एंड के लिए HTML, CSS और जावास्क्रिप्ट का उपयोग किया। हमने सिस्टम द्वारा उपयोग किए जाने वाले सभी डेटा को संग्रहीत करने के लिए MySQL का भी उपयोग किया। इससे समय बचाने, त्रुटियों को कम करने और कैंटीन के भीतर समग्र सेवा में सुधार करने में मदद मिलेगी।

कीवर्ड: कैंटीन प्रबंधन, वेब एप्लिकेशन, Django फ्रेमवर्क, ऑनलाइन फूड ऑर्डरिंग, डिजिटल बिलिंग, ऑर्डर ट्रेकिंग, MySQL

Table of Contents

ABSTRACT	iv
सार	v
CHAPTER: 1 : Introduction	1
1.1 Problem Identification	2
1.2 Information Gathering Tool	2
1.2.1 Survey Form	3
1.2.2 Interview	6
1.3 Hardware and Software Specification	7
1.3.1 Hardware Specification	7
1.3.2 Software Specification	7
CHAPTER: 2 SYSTEM ANALYSIS	9
2.1 Problem Analysis:	9
2.2 Feasibility Study	10
2.2.1 Economical Feasibility	10
2.2.2 Technical Feasibility	11
2.2.3 Behavioral Feasibility	11
2.3 Data Flow Diagram	12
2.3.1 DFD Level 0 Diagram	12
2.3.2 DFD Level 1 Diagram	13
CHAPTER: 3 : SYSTEM DESIGN	15
3.1 Structure Chart	15
3.2 Entity Relationship Diagram	17
3.3 UML Diagram	18
CHAPTER: 4 : TESTING	19
4.1 Unit Testing	19
4.2 Usability Testing	21

4.3 System Testing	21
CHAPTER: 5 IMPLEMENTATION	22
5.1 Visual Studio Code.....	22
5.2 Installing Python and Django	22
5.3 Setting Up Your Project	22
5.4 Configuring the Database	23
5.5 Project Implementation	23
5.6 Running the Project	23
5.7 Result.....	24
CHAPTER: 6 SAMPLE FORM	25
6.1 Sign in Page.....	25
6.2 Sign up Page.....	25
6.3 Home	26
CHAPTER: 7 CONCLUSIONS AND FUTURE SCOPE	27
7.1 CONCLUSIONS.....	27
7.2 Future Scope.....	27
CHAPTER: 8 REFERENCE.....	29

CHAPTER: 1 : Introduction

When you go into almost any college canteen during lunchtime, you will see a scene similar to this one. There are many students at the counter waiting for their turn, and workers are trying to fulfil numerous orders at once, which contributes to delays in students getting their orders – or even in some cases missing them altogether.

This issue is not just limited to one college; it applies to many canteens that still utilize manual processes of taking orders verbally or recording them on paper, thus creating confusion and wasting time for both students and staff members.

The inception of the project idea was born from observing actual situations like this. The intent was to make the process as simple as possible without creating excessive complexity; therefore students should have the ability to choose food, make an order, and determine when they can expect to receive their order without having to wait in a lengthy queue.

To achieve this end, we have created a Design of Canteen Management System that is online. This will allow students to log on to their own device, view the menu, and order food from that device; then their order will be directly placed in the kitchen in the order it was placed, thus eliminating confusion.

By also managing the food items within the database, the admin side of the system is focused on tracking orders as well as managing menu items.

1.1 Problem Identification

- a) The way orders are taken at the register (like speaking them or writing them on paper) can lead to a lot of errors. When multiple staff members serve 10 people at once, there is no way for them to accurately keep track of what's ordered if the order is taken either verbally or on a piece of paper.
- b) Because of the input of large groups at lunchtimes, there is congestion at the counter. Students have a lot of difficulty getting their orders placed due to the 30 minutes that they are given for their break; most of them take the majority of that time just standing in line waiting to order, and thus have no time to eat.
- c) There is no way to determine where an order is after placing it. A student cannot tell if an order is being worked on, is ready, or didn't get made because it was forgotten during prayer times.
- d) Paying cash for food slows the line down since the cashier has to count out change; and when the change is not available, there will be a lot back and forths, which slows down the line even further. Each day cash validation is very difficult since there is no written confirmation of how much money came in and went out.
- e) The boards used for displaying the prices and items for sale are manually changed either by erasing and writing, and this process takes a lot of time, is subject to change, and does not create an accurate written record of what changes were made.

1.2 Information Gathering Tool

Information gathering tools are methods to collect all the information about the existing system and user needs before developing a project.

1.2.1 Survey Form

DESIGN CANTEEN MANAGEMENT SYSTEM

SURVEY FORM

suryapratap245868@gmail.com [Switch account](#)

Not shared

Describe your overall experience with the current college canteen system, especially during busy hours.

Your answer

What problems are faced while ordering and receiving food in the canteen?

Your answer

How does waiting time in the canteen affect classes and daily routine?

Your answer

Explain difficulties faced during billing or payment in the current system.

Your answer

How can an online canteen management system improve the current situation?

Your answer

What features are expected in an online canteen management system?

Your answer

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Google Forms

RESPONSES

Describe your overall experience with the current college canteen system, especially during busy hours.

3 responses

✦ Summarise responses

Waiting time reduces break time and causes students to rush to classes.

Students experience long waiting time and confusion near the counter during peak hours.

The manual system makes order management difficult when many students are present

What problems are faced while ordering and receiving food in the canteen?

3 responses

✦ Summarise responses

Cash payment sometimes causes delay due to lack of change.

Sometimes wrong food items are received due to manual order handling

Slow service and lack of order organization create inconvenience

How does waiting time in the canteen affect classes and daily routine?

3 responses

✦ Summarise responses

Online ordering can reduce crowd and waiting time.

Some students become late for lectures due to long queues

Time management becomes difficult during short breaks

Explain difficulties faced during billing or payment in the current system.

3 responses

◆ Summarise responses

Cash payment sometimes causes delay due to lack of change

Manual billing may lead to calculation mistakes

Lack of digital payment options may slow transactions.

How can an online canteen management system improve the current situation?

3 responses

◆ Summarise responses

Online ordering can reduce crowd and waiting time

Order tracking can help students know when food is ready

Digital management can improve order accuracy

What features are expected in an online canteen management system?

3 responses

◆ Summarise responses

Online food ordering and menu display

Digital payment and order tracking facility

Updated menu availability and faster service

1.2.2 Interview

Interviewees Name – Naman, Aman

Contact No – 7582936578

Venue – College

Timing - 01:00 PM

Interview Naman (Student)

Siddharth: How does your current ordering experience in the canteen compare with previous years?

Naman: Honestly, it feels more like chaos than anything else. There is no scheduled system for us to order during lunch, and instead it's just a bunch of people standing around the counter waiting for someone to call their name. I have even missed classes because I spend the majority of my lunch waiting in line. Sometimes, I have to go back to the class because they gave me the wrong item.

Siddharth: How does the amount of time to order food affect your ability to study/relax after classes?

Naman: It directly decreases the amount of time I get to study/relax each day. On days where I have back to back classes, I often skip the canteen because it is not worth my time.

Siddharth: What would be a reasonable request for improvements to the canteen ordering system?

Naman: I would prefer able to select what I want from my phone, confirm my order through my phone, and then just go to the canteen to pick it up when it is ready, so that I don't even have to go back to the counter after my food is prepared.

Interview with Aman (Canteen Manager)

Surya: What difficulties do you have managing your orders by hand?

Aman: It is almost impossible during rush hour! With a limited amount of staff, we will be taking orders, cooking, billing and processing cash all at the same time. The combination of constantly taking orders and simultaneously working on many other tasks causes orders to get mixed up, disagreements between customers and staff, and a lot of added stress to everyone involved leading to more mistakes being made.

Surya: Are there issues related to record keeping due to the use of manual billing?

Aman: Yes, everyday! The cash count rarely matches the mental count at the end of the day. We have no system to audit the orders taken if we do not have the physical (written) tickets, and more times than not the tickets end up getting lost.

Surya: Would using technology help improve your operations?

Aman: Absolutely! If we could electronically take orders in the beginning and therefore have the kitchen staff just work their way through the queue of orders in order based on the order taken from the computer, everything would be a lot faster and easier to accomplish!

1.3 Hardware and Software Specification

1.3.1 Hardware Specification

Hardware Requirement: Efficient performance of the Design of Canteen Management System is dependent on proper hardware for both the system's development and execution. The system is developed and tested on an Intel Core i5 processor-based computer. The Intel i5 gives enough processing power, speed and performance to perform all tasks without any delay.

The memory in the system is 12 GB of RAM, which allows many operations to be performing simultaneously without lessening performance due to heavy load. The additional capacity of the RAM will enable to provide the necessary performance through as a result of using all of the additional memory.

The system uses a 512 GB SSD (solid-state drive) for data storage. The advantage of using the solid state drive is that it allows for quicker access to the data by having limited time required to load the data in comparison to a regular hard disk drive, or HDD. This in turn enhances the overall responsiveness of the system because the system has to provide timely access to the hardware for reading and writing data.

The projects have been developed and executed on Windows 11 (64-bit), which has been developed for modern development tools and thus has provided reliable and stable performance while building the project.

1.3.2 Software Specification

Operating System: The Windows 11 (64-bit) operating system is used for the system. This OS has reliable support for the tools required to develop the system and provides a stable environment for testing and development.

Programming Language: The system backend is programmed in Python version 3.x. Python is considered one of the easiest programming languages to learn because of its excellent library support and ease of integration with other components of web development.

Web Framework: The Django web framework is being used to simplify backend development. Django manages connections to the database, routing URL's, and provides built-in features such as user authentication and security.

Frontend Technologies: The front-end of the application consists of:

- a. HTML (Hypertext markup language), to create the structure and layout of web pages
- b. CSS (Cascading Style Sheets), to create the appearance of the front-end for a clean and responsive look
- c. JavaScript, to add interactivity and manage dynamic elements on web pages.

Database: The system makes use of a MySQL database to store information about users, menus, and order history. MySQL is a reliable, fast database that is well suited to use in web-based applications.

Code Editor: The development of the application is accomplished using Visual Studio Code (VS Code). This code editor has several helpful features such as syntax highlighting, debugging tools, and tools for organizing files in a project.

Web Browser: Google Chrome is used for running, testing, and debugging the web application.

CHAPTER: 2 SYSTEM ANALYSIS

The structured methodology of solving a problem includes breaking down a complicated process or information system into its various components in order to evaluate their effectiveness in achieving a particular end. This includes collecting and analysing user requirements in order to develop a system that works as efficiently as possible. System analysis will assist in determining what the new system should do and how it will function.

2.1 Problem Analysis:

Issue with Queueing Asynchronous Orders

There is no formal way for students to queue their food orders within the current Canteen System. During busy times, students are left waiting queuing for an uneven period with no clear idea of how many students are in front of them, resulting in some students not ordering at all.

Proposal for Solution

Students will be able to order digitally through their own device(s) allowing them to be placed into the digital queue from where it will be sent to the Canteen and processed in an orderly fashion, reducing pressure on the cash-register at the Canteen and providing an estimated wait time for their order.

Issue with Order History and Accountant Records

The Canteen staff have no efficient way of determining what has been ordered when & by how much it has cost the customer after the completion of a transaction. Reconciliation is done purely by memory and/or using freehand paper tickets.

Proposal for Solution

All orders completed through the System will be stored in the MySQL database. The Administrator will have the ability to view the Order History via a Dashboard, providing the ability to generate accurate Daily Reports and to view multiple statistical trends.

Issue with Absence Of Order Tracking

Currently, after placing an order; students do not have any method of knowing whether their order is being filled and/or if it has been neglected. This leads to students gathering at the counter asking about their orders which adds to the overall confusion.

Proposal for Solution

The system will let the users know in real-time via an on-screen display of the order status (i.e. Pending, Being Prepared and Ready). Students will be able to view their order from their computer and will not have to physically approach the counter.

Issue with Errors In Manual Invoicing/Dependence On Cash

Manual billing leaves room for arithmetic mistakes and exact cash requirements creates delays when there is no change for a customer; furthermore, there is no transaction trail for any disputed transaction.

Proposal for Solution

The system will automatically create a bill based on the order data. The system will include electronic methods of payment tracking and provide comprehensive transaction records that can be used for verification purposes.

2.2 Feasibility Study

A feasibility study is done to check whether the proposed system can be developed and used successfully.

2.2.1 Economical Feasibility

The total development cost has been estimated across two categories -

(a) Development cost (3 months)

S.NO	Specification	Cost (₹)
1	Backend & Database development	3,000
2	Frontend Design & documentation	3,000

Total = 6,000

(b) Other Expenses (3 months)

S.NO	Specification	Cost (₹)
1	Electricity charge	1,000
2	Documentation	400
3	Hardware cost	3,000
4	Software setup cost	2,000
5	Workplace Resources	500
6	System Processing Cost	1,500

Total = 8400 Rs

Total cost of whole project = 14,400 Rs

2.2.2 Technical Feasibility

a) Hardware Specifications

S. No	Specification	Description
1	Computer System	64-bit system
2	Storage	Min. 512 GB
3	RAM	8 GB recommended
4	Input Devices	Keyboard & mouse
5	Output Device	Monitor
6	Internet Connection	Required for system access

b) Software Specifications

S.NO	Specification	Description
1	OS	Windows 11
2	Browser	Chrome
3	Code Editor	VS Code
4	Frontend	HTML, CSS, JavaScript
5	Backend	Python and Django
6	Database	MySQL

2.2.3 Behavioral Feasibility

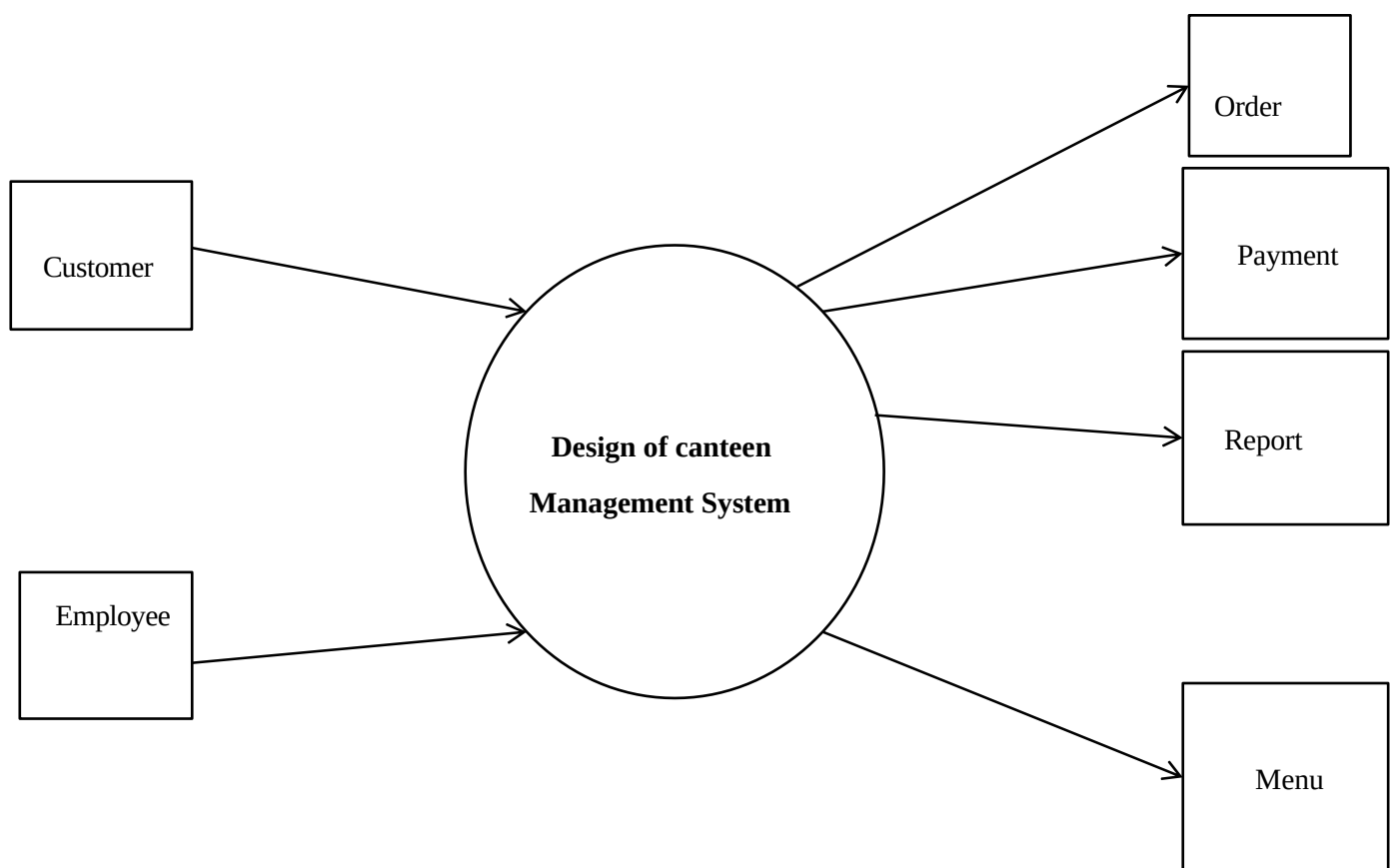
In behavioral feasibility, it is ensured that users can easily understand and efficiently use the proposed system.

- The system will provide clear instructions and proper guidance so that users can easily understand how to use the system without difficulty.
- The system will provide 24×7 service, allowing users to place orders or access information at any time without depending on canteen counter availability.

- c. We give a 1-year warranty and 6 months of free maintenance to fix any problems quickly.
- d. We provide a 1month training period to ensure users can confidently and effectively operate our system.
- e. We provide clear directions for users in case of system errors or uncertain and provide reinstallation between training period.

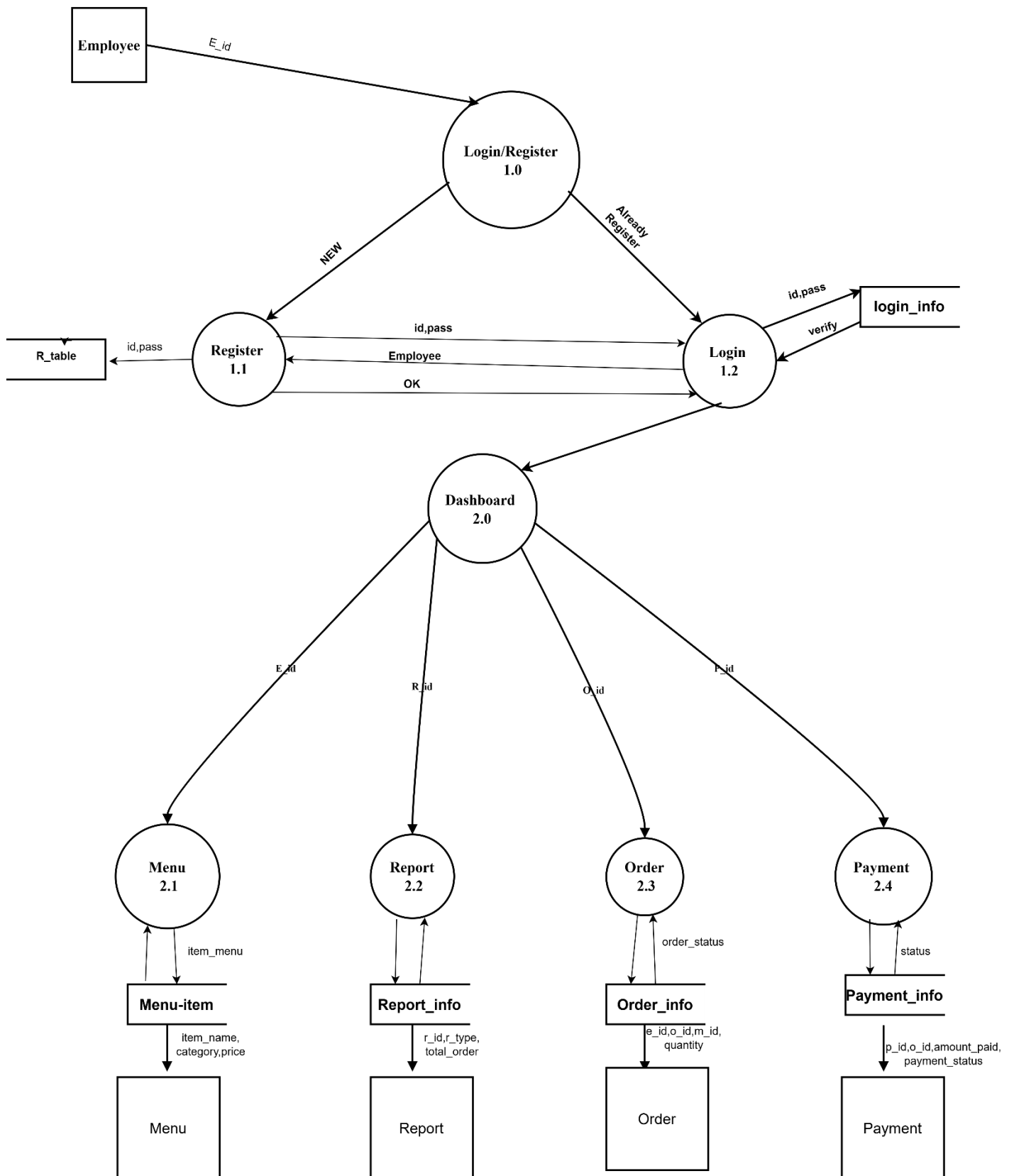
2.3 Data Flow Diagram

2.3.1 DFD Level 0 Diagram

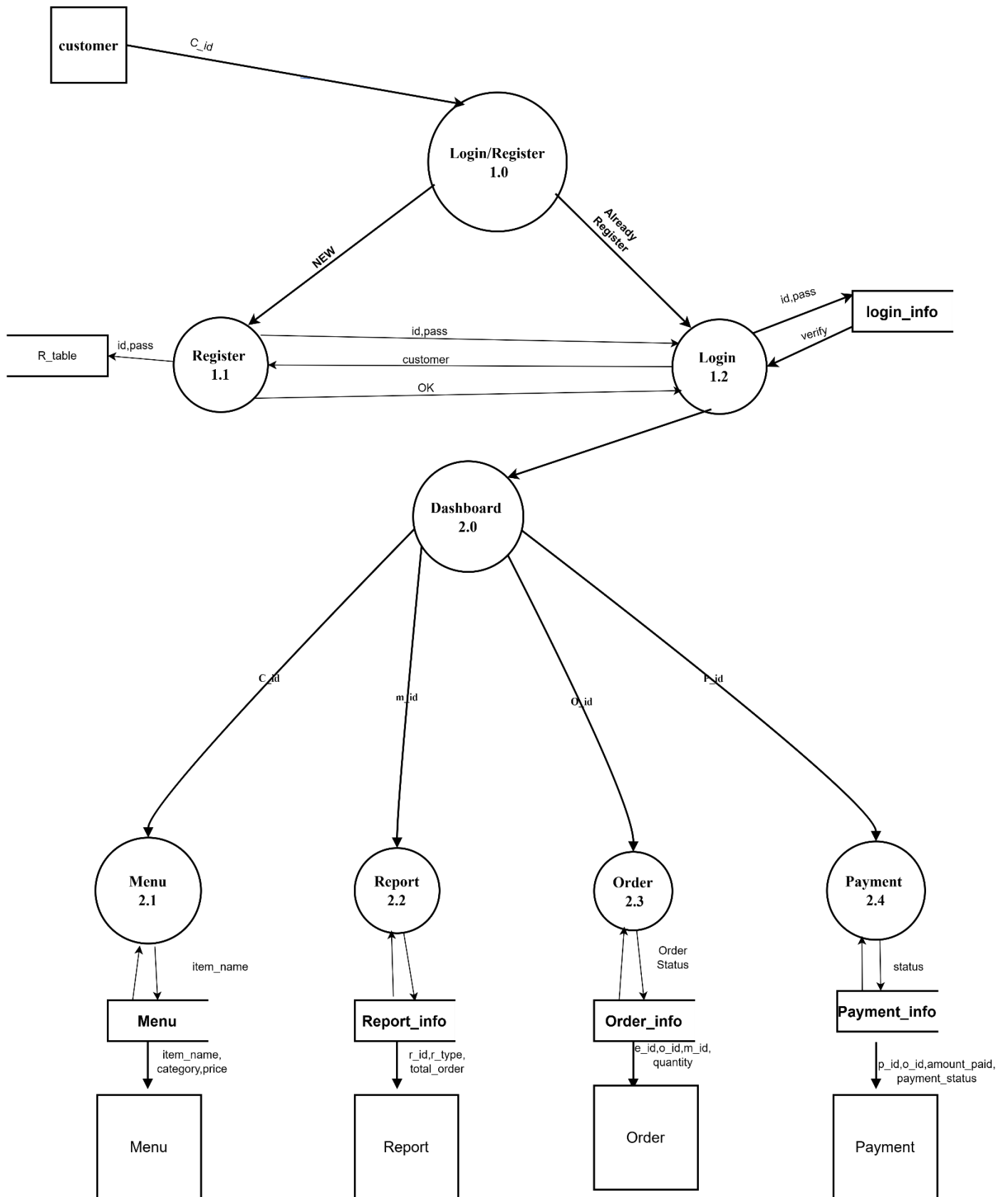


2.3.2 DFD Level 1 Diagram

a) Employee



b) Customer



CHAPTER: 3 : SYSTEM DESIGN

3.1 Structure Chart

The database consists of five core entities. Each table is designed to store a specific type of data, with primary and foreign keys establishing relationships between them.

a) Food

S. No	Attribute/ Field Name	Datatype	Size	Key
1	F_item_no	Integer	10	Primary Key
2	F_Name	Varchar	50	
3	F_type	Varchar	30	
4	F_Price	Decimal	10	

b) Customer

S. No	Attribute/ Field Name	Datatype	Size	Key
1	C_id	Integer	10	Primary Key
2	C_name	Varchar	50	
3	C_type	Varchar	20	
4	Phone_no	Integer	10	

c) Order

S. No	Attribute/ Field Name	Datatype	Size	Key
1	Order_id	Integer	10	Primary key
2	Order_no	Integer	10	
3	Order_date	Date	10	
4	C_id	Integer	10	Foreign Key (Customer)

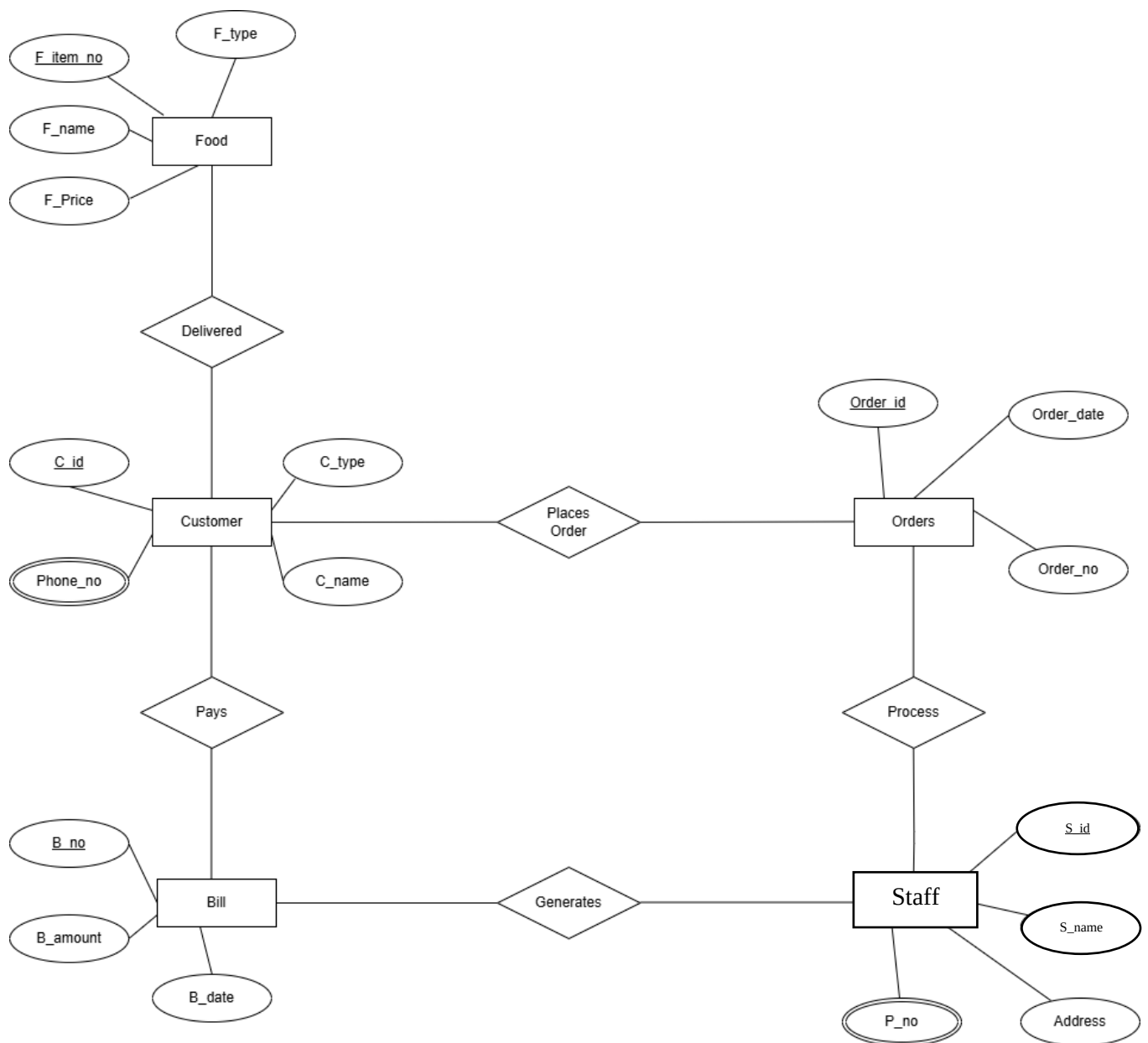
d) Bill

S. No	Attribute/ Field Name	Datatype	Size	Key
1	B_no	Integer	10	Primary key
2	B_amount	Decimal	10	
3	B_date	Date	10	
4	Order_id	Integer	10	Foreign Key (Order)
5	C_id	Integer	10	Foreign Key (Customer)

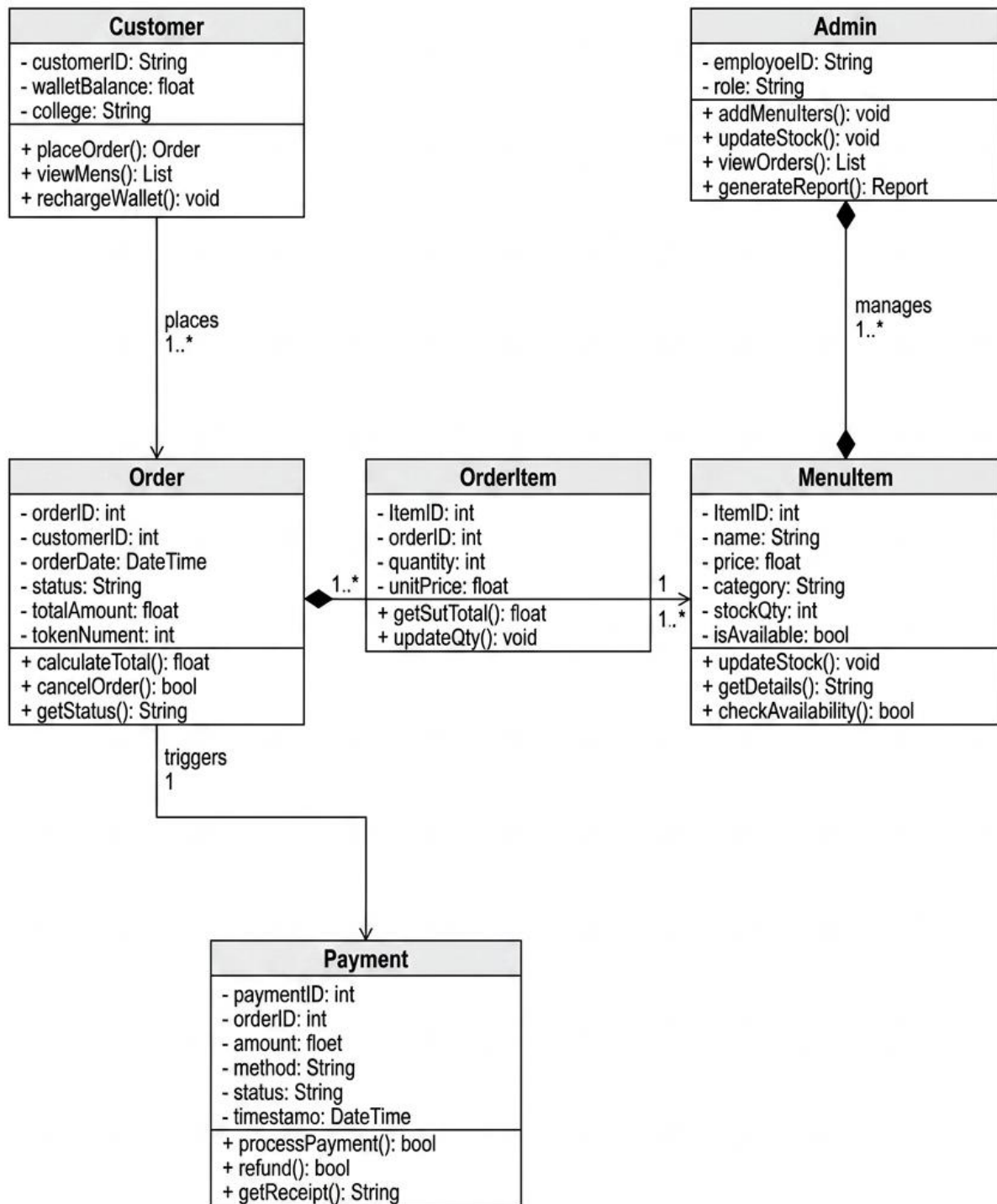
e) Administrator

S. No	Attribute/ Field Name	Datatype	Size	Key
1	S_id	Integer	10	Primary key
2	S_name	Varchar	50	
3	P_no	Integer	10	
4	Address	Varchar	100	

3.2 Entity Relationship Diagram



3.3 UML Diagram



CHAPTER: 4 : TESTING

4.1 Unit Testing

Unit testing targeted each module with both valid and invalid inputs. The objective was to confirm that the system behaves correctly in all edge cases, not just under ideal conditions.

Test Case ID	Section	Element	Test Data	Expected Result	Actual Result
1	Admin Login	Username, Password	No Data	Show Fill all Field	Passed
			admin@123 , wrong	Invalid Credentials	Passed
			admin@canteen , 1234	Login Successful	Passed
2	User Login	Email, Password	Empty Feilds	Show Error	Passed
			naman@gmail , 1234	Invalid Email	Passed
			naman@gmail.com , 1234	Login Successful	Passed
3	Registration	Name, Email, Phone	Empty data	Error Message	Passed

			Surya,25ca1su@.com, 7582936578	Invalid Detail	Passed
			Surya, 25ca1su50@gmail.com , 7582936578	Successfully Registered	Passed
4	Food Ordering	Item Quantity	No Item Selected	Show Warning	Passed
			-1	Invalid Input	Passed
			1	Order Placed	Passed
5	Payment	Payment Method	No Selection	Show Error	
			(abc@upi)	Failed Transaction	Passed
			Cash payment	Payment Successfully	Passed

4.2 Usability Testing

Usability assessments were conducted through the observation of users who had not been previously involved in the creation of a particular system. The observers were able to assist in determining how the users were able to navigate through the system with no instructions and to note any points of confusion or hesitation.

All new users were able to successfully create an account without assistance; the account creation form was easy to read due to the form being labelled well, and the error messages identified exactly which fields needed correction to continue the process. The observers found that the menu page was very quick to navigate, and food categories and prices were easy to find.

The checkout process was observed to have 3 steps (review shopping cart, select payment method, confirm order), all users indicated that they viewed the checkout process as an acceptable experience.

The most informative comments provided by users who provided feedback, the status of their order (Pending, In Preparation, Ready) were all easily described and therefore eliminated the need for an additional description to identify what the status meant. One user requested an estimated time of when the order status would be completed, this will be included on the list of future enhancements to the system. Overall, all observers noted that the system was easy to navigate.

4.3 System Testing

The system was tested throughout all modules as a whole to ensure that the data flowed seamlessly between each module without interruption and was consistent across all modules. All browsers, including Google Chrome and Microsoft Edge, were tested. No compatibility issues existed among browser vendors. I also tested this application through an Android smartphone and mid-range tablet, to ensure responsive CSS functionalities on devices with smaller screen sizes.

I validated that all database transactions completed correctly (creation of an order, status change on orders, bills created for orders, and retrieval of an order) during each phase of the order lifecycle, and the database (MySQL) tracked all updates correctly. I was also able to verify that the application would not permit the creation of an order for an item not currently available when the order was sent to the database layer.

Through informal load testing, I created multiple, quick succession orders to validate the application's performance under stress. The application used the Django ORM to properly insert multiple concurrently created orders into the database without data integrity issues or errors, which should support the application under anticipated user loads in a university cafeteria setting.

Based on the results of integrated testing, the system meets all functional requirements as defined in the analysis phase of testing, and there were no critical defects found during testing.

CHAPTER: 5 IMPLEMENTATION

The purpose of this section is to provide a detailed guide on setting up a development environment and deploying a Canteen Management system using all open-source and free software.

5.1 Visual Studio Code

- a) In your browser go to <https://code.visualstudio.com/>.
- b) On the website select the correct installer for your operating system (Windows, macOS or Linux).
- c) Run the installer that you downloaded. Follow the steps in the setup wizard and accept the license agreement.
- d) Open Visual Studio Code and go to the Extensions pane (Ctrl + Shift + X) and install the following extensions: Python, Django, Live Server
- e) In Visual Studio Code settings and confirm that the Python interpreter path is correctly set (Python: Select Interpreter).

5.2 Installing Python and Django

- a) Go to <https://www.python.org/downloads/> and download the latest stable version of Python (3.x).
- b) During the Python installation, make sure you check the box "Add Python to PATH", as this is required to run Python from the CLI.
- c) To verify that Python has been installed correctly open CMD and type `python --version`.
- d) Use pip to install Django with: `pip install django`
- e) Verify that Django has installed correctly with: `django-admin --version`

5.3 Setting Up Your Project

- a) Open a new terminal in your work area.
- b) Create a new Django project with the command: `django-admin startproject canteen_project`.
- c) Change into the project directory by running: `cd canteen_project`.
- d) Use the command `python manage.py startapp canteen_app` to create an application module for your project.
- e) Add `canteen_app` to `INSTALLED_APPS` in `settings.py`.
- f) Start a local development database server to test the project with: `python manage.py runserver`.
- g) Open a web browser and type in `http://127.0.0.1:8000/` to check that the project is up and running.

5.4 Configuring the Database

- a) By default, Django will use SQLite as its default database. To run your production-level application using a MySQL database, you must modify the DATABASES section of your settings.py with the appropriate host, database, username and password information.
- b) Inside canteen_app/models.py, define your models for each of the following entities: User, MenuItem, Order, OrderItem, and Bill.
- c) Generate the migration script files by running: `python manage.py makemigrations`.
- d) Create the actual database tables using the generated migration scripts by running: `python manage.py migrate`.
- e) Create a super user for the admin panel by running: `python manage.py createsuperuser`.
- f) Access the admin panel at <http://127.0.0.1:8000/admin/>.

5.5 Project Implementation

Using Django's Model View Template (MVT) makes it easy to maintain a well-organized design and keep functionality separate from presentation.

- a. **MODELS (models.py):** The model represents the table(s) in the database and keeps track of how the tables relate to one another. If the model changes, you can use Django's migration framework (to automatically update the database).
- b. **VIEWS (views.py):** The view contains the business logic (programming that runs when a user clicks) that will handle the user's request via HTTP. Each view method gets the HTTP request, processes it (including any data retrieval or updates to the database), and produces a rendered HTTP response by calling the appropriate view template.
- c. **TEMPLATES (templates/):** Templates are HTML files that describe how users will see their screen when they view your application. When you render an HTML page, the template engine will dynamically insert all of the data (like menu options and order status) into the HTML.
- d. **URLs (urls.py):** URLs are used to map the browser's address to the appropriate view. When a user requests a URL, the appropriate view method will execute and render the HTML page.

Some of the main modules are: User registration and login; display of menu items and ability to filter by category; shopping cart and order placement; dashboard showing the status of orders; billing and payments; and Admin panel for managing menus and orders.

5.6 Running the Project

- a. Use the terminal to navigate the canteen_project directory
- b. Execute - `python manage.py runserver`
- c. Open a web browser and enter - `http://127.0.0.1:8000`

- d. The available food items appear on the homepage for the canteen with a search function
- e. New user clicks on register (will register them as a new user), provides required information (this creates a user in the database) and then they can continue to the main menu
- f. Existing user logs into the canteen app and is taken directly to the user's order dashboard
- g. To access through Administrator, go to <http://127.0.0.1:8000/admin/> and login with superuser user details.

5.7 Result

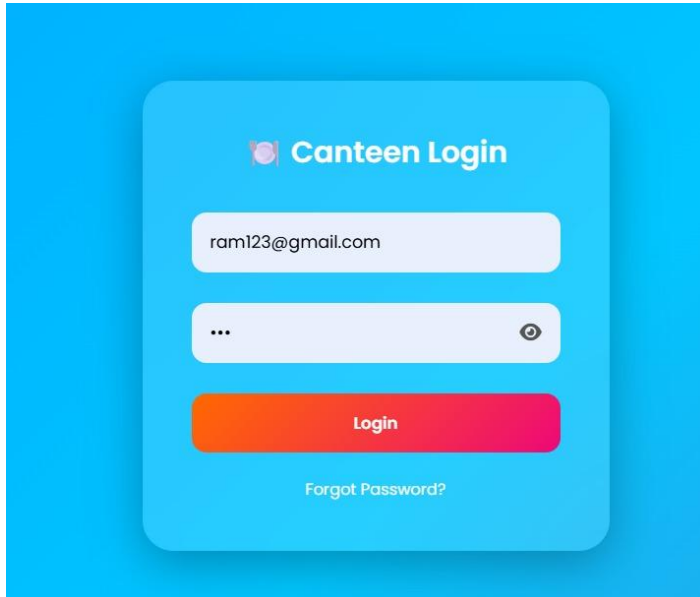
We built, tested and validated the entire Design of Canteen Management System in our local dev env and all functionality completed successful test validations without any critical errors across all modules (registration, login, menu browse, order, payment, and admin).

During final testing we tested on three different devices (desktop, laptop and mobile) and two different browsers (Chrome and Edge). All devices produced consistent data/results with no issues.

From multiple test sessions, the database stored and was able to retrieve all records accurately and was not able to observe any data loss or corruption.

CHAPTER: 6 SAMPLE FORM

6.1 Sign in Page



A login form titled "Canteen Login" with a blue background. It features a light blue rounded rectangle containing the title, an email input field with "ram123@gmail.com", a password input field with a toggle icon, a red "Login" button, and a "Forgot Password?" link.

Canteen Login

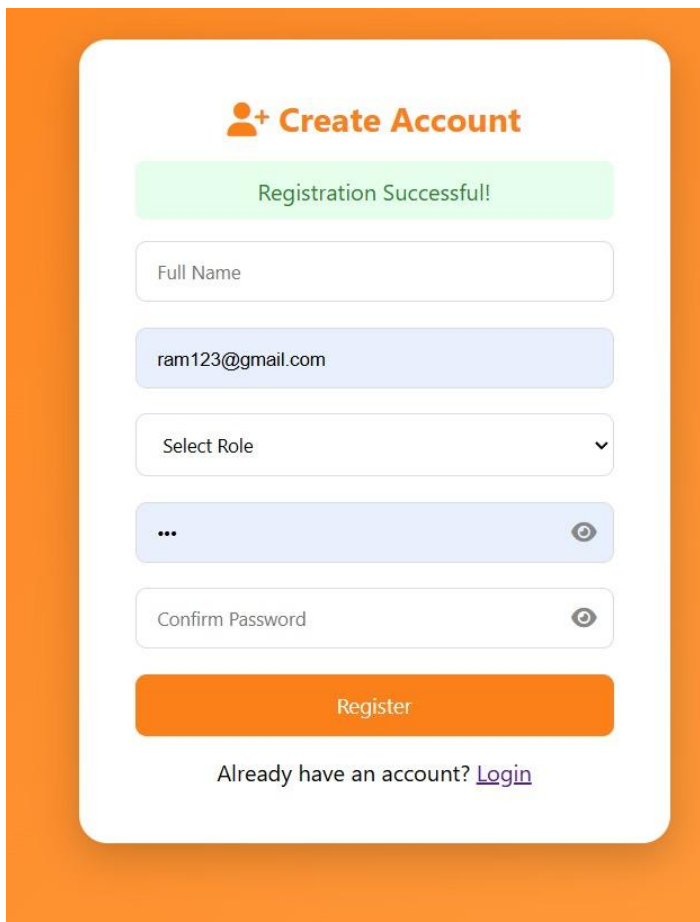
ram123@gmail.com

...

Login

[Forgot Password?](#)

6.2 Sign up Page



A registration form titled "Create Account" with an orange background. It features a white rounded rectangle containing a success message, a "Full Name" input field, an email input field with "ram123@gmail.com", a "Select Role" dropdown, a password input field with a toggle icon, a "Confirm Password" input field with a toggle icon, an orange "Register" button, and a link to "Login" if the user already has an account.

Create Account

Registration Successful!

Full Name

ram123@gmail.com

Select Role

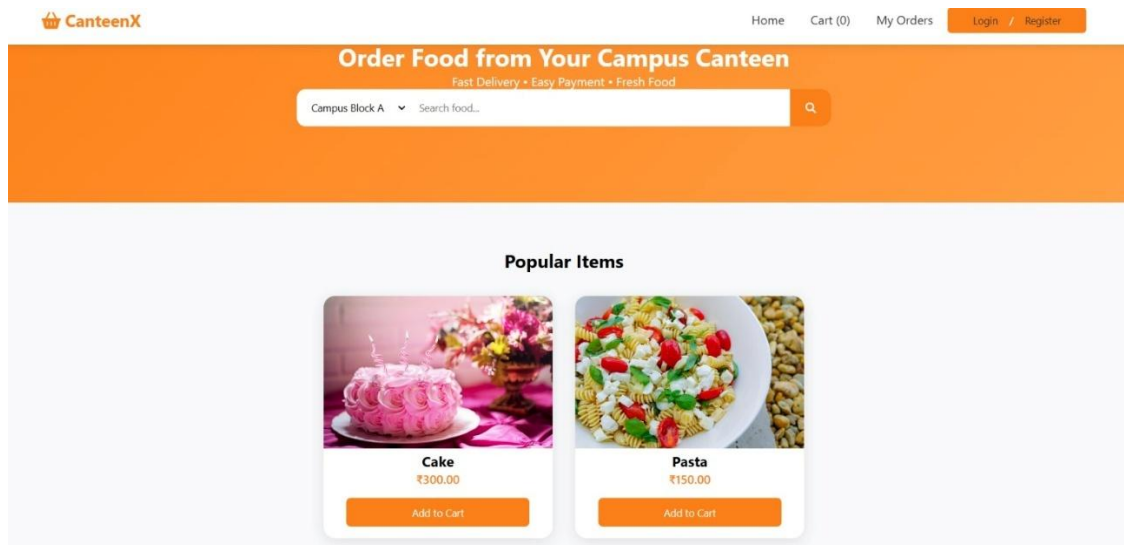
...

Confirm Password

Register

Already have an account? [Login](#)

6.3 Home



CHAPTER: 7 CONCLUSIONS AND FUTURE SCOPE

7.1 CONCLUSIONS

The Design of Canteen Management System successfully resolves the operational challenges that led to its formation. Previously a disorganized, mistake-laden procedure reliant upon manual input, has transitioned into a digital system where there is a separately identified participant (student, kitchen employee and administrators) who can perform their roles effectively utilizing the designated interface.

For students, the long wait in line has been replaced by being able to see a current menu and purchase via the use of their smartphone, computer or any other connected device. With regular updates related to purchase status being communicated directly with the student ordering the food eliminates the need for students to wait in line, thus reducing the propensity to congregate at the counter.

When employees of the canteen receive orders, the orders come to them as structured versus unstructured data, creating an established queue through which they will process the remaining orders. Administrators are also able to complete menu changes instantaneously, review the day's transaction information, and examine customer behaviours, which was not achievable under the previous system.

The technology components used to implement the Canteen Management System (Django for the application server, MySQL as the Database Server, HTML, CSS and JavaScript as the front-end development tools) were purposely selected to ensure long-term usability, lowest total cost, and minimal maintenance. All have been designed so they will run on relatively lower-end hardware and do not require any special software.

7.2 Future Scope

The following are a number of improvements that were discovered during the design, development and testing phases that will increase the effective capabilities of the application in future releases:

- a. **UPI and Card Payment Integration:** Currently, payment tracking is supported but not live, digital transactions. By providing an interface to payment gateways (e.g., Razorpay, Paytm) cashless ordering can be completed from order placement to delivery;
- b. **Push Notifications:** A notification system (via browser and SMS) will allow students to be alerted anytime there is a change in the status of their order from In Preparation to Ready, so they will no longer need to check the screen after placing their orders;
- c. **Personalized Recommendations:** By accessing historic orders placed by a student, a simple recommendation engine could suggest items based upon individual likes and pattern orders based upon time of day;

- d. Native Mobile Application:** Providing a native Android and iOS application would give students who access the application mostly through their smart phones, better access and use build browsing capability while offline;
- e. Multi-Canteen Capabilities:** The application architecture could easily be configured to accommodate multiple canteens on one campus, each able to maintain their own menu, staff log in, and reporting dashboard;
- f. Increased Security:** By implementing OTP-based logins, two-factor authentication and encrypting storage of payment related fields, the application will maximize its security in preparation for moving into production; and
- g. Analytics Dashboard:** There will be a separate admin facing analytics page to assist with monitoring effectiveness of the system.

CHAPTER: 8 REFERENCE

<https://www.w3schools.com/>

<https://www.geeksforgeeks.org/>

<https://www.tutorialspoint.com/>

<https://developer.mozilla.org/>

<https://stackoverflow.com/>

<https://www.freecodecamp.org/>

<https://github.com/>

<https://jquery.com/>

<https://getbootstrap.com/>

<https://www.mysql.com/>

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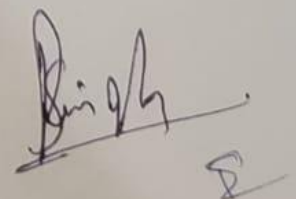
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